SENDAZA LIQUIDITY ENGINE

Sprint 1: Exact Amounts and Registries

How SLE learned what it supports without losing financial precision

Delivered

Before and Goal

Sprint 0 supplied a runnable workspace, but it did not understand money, assets, networks, or markets. Sprint 1 gave SLE an authoritative dictionary of supported configurations and a safe ruler for measuring amounts.

> **Core idea:** asset and network are separate. USDT is an asset; Ethereum, Tron, and Solana are networks. USDT on each network has separate addresses, fees, custody routes, and inventory.

Data Model

Entity	Example and purpose
Asset	USDT: identifies crypto value and precision.
Fiat	NGN: identifies fiat denomination and precision.
Network	Ethereum: defines blockchain behavior.
Asset-network	USDT/Ethereum: defines the token route and capabilities.
Market	USDT/NGN: defines a supported purchase market.

Exact Amount Flow

```
"1000.01" NGN -> validate 2 decimals -> 100001n
-> PostgreSQL BIGINT -> format exactly -> "1000.01"
```

Money never uses JavaScript floating-point `Number`. Excess precision is rejected instead of silently rounded.

What Was Built

- Prisma models and PostgreSQL constraints for registries and configuration versions.

- Atomic-unit conversion using `bigint`.
- A repository boundary between domain services and Prisma.
- Database-backed `GET /assets` and `GET /markets`.
- Illustrative multi-network USDT routes proving the model is network-aware.

System Link

Quotes use markets and precision. Treasury tracks inventory by asset-network. The fee engine prices networks separately. Fireblocks and withdrawals select a configured route, never only an asset symbol.

Evidence and Limits

Tests covered exact conversion, excess precision, registry validation, and mapping. Migrations ran on Railway PostgreSQL, and a live query returned an asset, two USDT routes, and a USDT/NGN market. This sprint did not fetch prices, calculate fees, reserve inventory, or move funds.

Next Sprint

Sprint 2 protects these endpoints by authenticating Sendaza, preventing replay, enforcing idempotency, and delivering committed events durably.

Glossary

Atomic unit: smallest exact unit. **Precision:** permitted decimal places. **Registry:** authoritative supported configuration. **Constraint:** a database rule rejecting invalid data.